

Subnet Mask and Wildcard Mask

- **192.168.10.0 / 30**

In here 30 means number of network bits. This belongs to Class C by default. Therefore it should have only 24 network bits. But there are 30 bits for network. That means 6 bits were borrowed from the host part to network part.

Network bits = 30 Host bits = 2

11111111 11111111 11111111 **111111**00

(In here, 6 bits are borrowed from last octet to network bits. Then total is 30 bits for network part and 2 bits for host part.)

Number of subnets: $2^6 = 64$ subnets

Number of hosts: $2^2 - 2 = 2$ hosts per network.

Then subnet mask is decimal format of this.

11111111 11111111 11111111 11111100
255. 255. 255. 252

Just like a subnet mask, a wildcard mask is 32 bits long. It acts as an inverted subnet masks. In wildcard mask,

1. **The zero bits indicate that the corresponding bit position must match the same bit position in the IP address.**
2. **The one bits indicate that the corresponding bit position doesn't have to match the bit position in the IP address.**

Subnet Mask – 11111111 11111111 11111111 11111100 → 255.255.255.252

Wild card Mask – 00000000 00000000 00000000 00000011 → 0.0.0.3

- **125.1.0.0 / 16**

In here 16 means number of network bits. This belongs to Class A by default. Therefore it should have only 8 network bits. But there are 16 bits for network. That means 8 bits were borrowed from the host part to network part.

11111111 11111111 00000000 00000000

Network bits = 16 Host bits = 16

(In here, 8 bits are borrowed from second octet to network bits. Then total is 16 bits for network part and 16 bits for host part.)

Number of subnets: $2^8 = 256$ subnets

Number of hosts: $2^{16} - 2 = 65534$ hosts per network.

Subnet Mask – 11111111 11111111 00000000 00000000 → 255.255.0.0

Wildcard Mask – 00000000 00000000 11111111 11111111 → 0.0.255.255

- **10.0.0.0 / 8**

In here 8 means number of network bits. This belongs to Class A by default. Therefore it should have only 8 network bits. There are no bits borrowing from the host.

11111111 00000000 00000000 00000000

Network bits = 8 Host bits = 24

(In here, no bits are borrowed.)

Number of subnets: $2^0 = 1$ subnet.

Number of hosts: $2^{24} - 2 = 16777214$ hosts per network.

Subnet Mask – 11111111 00000000 00000000 00000000 → 255.0.0.0

Wildcard Mask – 00000000 11111111 11111111 11111111 → 0.255.255.255

- **172.20.0.0 / 16**

In here 16 means number of network bits. This belongs to Class B by default. Therefore it should have only 16 network bits. There are no bits borrowing from the host.

Network bits = 16 Host bits = 16
(In here, no bits are borrowed.)

Number of subnets: $2^0 = 1$ subnet.

Number of hosts: $2^{16} - 2 = 65534$ hosts per network.

Subnet Mask – 11111111 11111111 00000000 00000000 → 255.255.0.0

Wildcard Mask – 00000000 00000000 11111111 11111111 → 0.0.255.255

- **122.31.0.0 / 16**

In here 16 means number of network bits. This belongs to Class A by default. Therefore it should have only 8 network bits. But there are 16 bits for network. That means 8 bits were borrowed from the host part to network part.

11111111 11111111 00000000 00000000

Network bits = 16 Host bits = 16

(In here, 8 bits are borrowed from second octet to network bits. Then total is 16 bits for network part and 16 bits for host part.)

Number of subnets: $2^8 = 256$ subnets

Number of hosts: $2^{16} - 2 = 65534$ hosts per network.

Subnet Mask – 11111111 11111111 00000000 00000000 → 255.255.0.0

Wildcard Mask – 00000000 00000000 11111111 11111111 → 0.0.255.255